System and Method for Structured Cognitive-State Modeling Using a Fixed Slot Architecture, Slot-Level Domain-Descent Attention, Structured Supervision, and Engine-Aligned Evaluation

Cross-Reference to Related Applications

None.

Federal Sponsorship

No federal sponsorship.

Field

The invention relates to computer-implemented neural architectures, representation learning systems, and machine-assisted evaluation systems for sequence-processing models, including large language models. More particularly, it relates to fixed-slot latent cognitive-state modeling, slot-level attention with domain-ordered constraints, structured supervision from cognitive labels, prerequisite-aware gating, and engine-aligned scoring and evaluation of neural outputs.

Background

Conventional transformer architectures operate primarily over token positions and generic hidden states. They do not natively preserve a fixed structured latent bank aligned to a preselected cognitive-state ontology, nor do they typically provide a direct path from that internal structure to a separate scoring or evaluation engine with auditable outputs.

Existing architectures therefore often lack one or more of the following technical properties:

1. A fixed latent slot bank that remains stable across variable-length token sequences.
2. Attention constraints applied over explicit latent slots rather than over raw token positions alone.
3. A structured supervision path that converts cognitive labels into a constrained target state.
4. A prerequisite-aware gating mechanism coupled to structured state relationships.
5. A reproducible evaluation bridge from neural outputs into a separate scoring engine.
6. A clean distinction between mathematically enforced constraints and heuristic neural approximations.

There is therefore a need for a neural architecture that can map variable-length sequences into a fixed structured state bank, process that state bank under slot-level ordering constraints, train against explicit structured targets rather than only next-token loss, emit outputs that can be evaluated by a separate scoring engine, and support auditable benchmark, validation, and calibration workflows.

Summary of the Invention

In one aspect, the invention is a neural architecture that combines token embeddings and positional representations, a cognitive-state branch that produces weights over a fixed slot bank, a slot-level domain-descent attention mechanism operating over a fixed set of slots, a prerequisite gate or RPM-related gate, additional auxiliary heads for inversion or proposition scoring, and an evaluation bridge that maps neural outputs into a separate scoring engine.

In another aspect, the invention uses a fixed latent state bank as an explicit intermediate representation rather than treating the structured state solely as a post hoc label.

In another aspect, the invention performs slot-level aggregation from token states into the fixed slot bank, applies attention and constraints over the slot bank, and projects the updated slot states back into token-level representations.

In another aspect, the invention uses structured labels to form target distributions on an open simplex and trains the model against those targets.

In another aspect, the invention evaluates model outputs by converting the learned cognitive-state representation into inputs for a separate scoring engine, thereby keeping the model architecture and the scoring engine distinct while still coupled through a defined interface.

Brief Description of the Drawings

Figure 1 illustrates an overall architecture including a token embedding path, positional encoding, cognitive-state branch, repeated CAS layers, output heads, and external evaluation path.

Figure 2 illustrates a fixed slot state bank showing token-to-slot weights, slot aggregation, multiple domain bands of slots, and projection back to token positions.

Figure 3 illustrates slot-level domain-descent attention showing slot queries, keys, and values, a domain-ordering mask, an optional feedback path, and masked-token handling.

Figure 4 illustrates a prerequisite gate or RPM path showing hidden-state input, gate computation, satisfaction signals, and gated output.

Figure 5 illustrates a structured supervision path showing labeled cognitive scores, target state distribution generation, open-simplex normalization, and cognitive metric loss.

Figure 6 illustrates an evaluation bridge showing model output, state renormalization, a PyO3 bridge, a Rust engine scoring path, and inversion or composite outputs.

Detailed Description

1. System Context

The system may be used as a standalone model, as a research architecture, or as part of a broader evaluation service.

The system may include model assembly, slot-level domain-descent attention, a prerequisite gate, auxiliary heads, an open-simplex embedding path, a cognitive metric loss path, a training loop, an evaluation bridge, and a separate state or scoring engine.

2. Exemplary Inputs

The system may accept one or more of the following:

- tokenized text sequences;
- token masks identifying valid and padded positions;
- structured cognitive labels or cognitive scores;
- prompts or responses for evaluation mode; and
- model configuration values such as slot count, head count, and hidden dimension.

In one embodiment, the system processes natural-language sequences while preserving a structured fixed latent state bank.

3. Fixed Slot Architecture

In one embodiment, the architecture maintains a fixed latent state bank. In a representative embodiment, the slot bank contains forty slots grouped into multiple domain bands.

The slot bank is not merely a post-processing label space. It acts as an explicit intermediate representation. Token states are mapped into slot weights, slot states are aggregated, transformed, and then projected back into token-level representations.

The fixed slot bank provides a stable internal structure across variable-length sequences.

4. Slot-Level Domain-Descent Attention

In one embodiment, attention is performed over the fixed slot bank rather than solely over token positions.

The process may include:

1. receiving token hidden states and token-to-slot weights;
2. aggregating token representations into slot states;
3. applying slot-level attention subject to a domain-ordered mask;
4. optionally allowing a feedback path across designated domain bands; and
5. projecting the updated slot states back onto token positions using the slot weights.

This produces a structurally constrained pathway that differs from conventional token-only attention.

In one embodiment, padding-mask handling is threaded through the relevant modules so padded token positions do not materially alter unmasked outputs.

5. Prerequisite Gate

In one embodiment, a prerequisite gate receives hidden states and computes one or more satisfaction values that gate subsequent activation flow.

The gate can be used to reflect prerequisite relationships or RPM-related integrity structure. The gate may return both gated outputs and satisfaction values.

This gate may be combined with the slot-level attention path and with auxiliary loss terms.

6. Structured Supervision

In one embodiment, structured cognitive labels are converted into target state distributions.

The process may include:

1. reading operator and domain labels from structured label inputs;
2. mapping those labels into a target state vector;
3. normalizing the target onto an open probability simplex; and
4. training the model against the predicted cognitive-state output using a cognitive metric loss.

This differs from generic next-token-only training because the model is supervised toward an internal structured target rather than solely toward surface text.

7. Open Simplex and Cognitive Metric Loss

In one embodiment, the model predicts cognitive-state vectors that lie on an open probability simplex.

An open-simplex projection may be used in the embedding path and a Fisher-Rao-based cognitive metric loss may be used in the loss path. This provides a mathematically constrained target space for the structured state representation.

8. Auxiliary Heads and Architecture Extensions

The architecture may include auxiliary heads such as inversion detection, proposition or truth-value scoring, and language-model output heads.

Some embodiments may include heuristic neural approximations in addition to more formal constraint paths.

9. Engine-Aligned Evaluation

In one embodiment, model outputs are evaluated through a separate scoring engine.

The process may include:

1. extracting the model's predicted cognitive-state vector;
2. renormalizing it to satisfy engine-side validation requirements;
3. passing it through a bridge into a scoring engine;
4. deriving operator or domain scores, inversion outputs, and composite scoring from the engine; and
5. returning those outputs as part of a benchmark or evaluation report.

This produces a separation between the learned model and the scoring engine while preserving a formal interface between them.

10. Optional Language Integration

In some embodiments, the architecture may be coupled to an executable language or compiler toolchain for generating, validating, replaying, or constraining structured examples.

11. Example Workflow

An example workflow may include:

1. receiving a tokenized prompt or response sequence;
2. embedding the sequence and positional information;
3. generating token-to-slot weights over a fixed slot state bank;
4. aggregating token states into slot states;
5. applying slot-level domain-descent attention;
6. applying prerequisite gating and any auxiliary heads;
7. producing a cognitive-state output on the open simplex;
8. converting that output into engine-compatible inputs; and
9. computing evaluation scores using the separate engine.

12. Alternative Embodiments

The invention is not limited to the exact implementation currently described. Alternative embodiments may include:

- slot counts other than forty;
- domain groupings other than four groups of ten;
- different slot-order masks or dependency structures;
- different prerequisite gate formulations;
- different cognitive metric losses;
- different auxiliary heads or proposition vocabularies;
- different bridges to scoring or validation engines; and
- non-language sequence domains.

13. Technical Benefits

Potential technical advantages include a stable structured internal representation across variable-length inputs, explicit slot-level constraints rather than implicit token-only biasing, a direct supervision path from structured labels to internal state, improved auditability between model state and scoring output, and reproducible evaluation through a separate engine.

These should be understood as technical advantages or design objectives unless and until validated performance evidence is added.